

MOBAN Chapter 6 — Comprehensive Study Guide

Broadband Access Networks (IPACT)

PART 1: THEORY SUMMARY

1. Physics & Generic Architecture

1.1 Role of the Access Network

The **access network** connects customer terminals (or customer premises networks) to the backbone/core network. It sits between the LAN and the backbone.

Two classic types of fixed access networks:

- **Twisted pair** — telephone network (PSTN), terminal = telephone, gateway = local exchange (LEX)
- **Coaxial cable** — CATV network, terminal = television, gateway = head end

A third type — **optical fiber** — is now increasingly important in the access network.

1.2 Physical Media

Medium	Structure	Typical bandwidth
Twisted pair	Two isolated copper wires twisted together; twisting reduces noise sensitivity	~10 MHz
Coaxial cable	Central copper wire + copper shield + insulator; excellent noise shielding	~1 GHz
Optical fiber	Glass wire with higher-refractive-index core; light guided by total internal reflection	~10 THz

Attenuation = signal power loss per km, expressed in dB/km:

$$\text{Attenuation (dB)} = 10 * \log_{10}(P_{\text{in}} / P_{\text{out}})$$

- e.g., 3 dB/km means power halved every km
- Optical fiber has the **lowest attenuation** in the infrared range (~1300-1550 nm)

1.3 Optical Fiber Physics

Key relation (wavelength, frequency, speed):

$$\lambda * f = v = c / \eta$$

Symbol	Meaning	Value
λ	Wavelength	—
f	Frequency	—
v	Speed in medium	—
c	Speed of light in vacuum	300,000 km/s
η	Refractive index	~1 for air; ~1.5 for optical fiber

Quick reference calculations:

- Speed in optical fiber: $v = 300,000 / 1.5 = \mathbf{200,000 \text{ km/s}}$
- RTT over 100 km fiber: $RTT = 2 \times 100 / 200,000 = \mathbf{1 \text{ ms}}$
- Infrared at 1500 nm: $f = 200,000 \times 10^6 / (1500 \times 10^{-9}) = \mathbf{133 \text{ THz}}$

1.4 Shannon Capacity

The maximum bitrate over a bandwidth-limited noisy channel:

$$C = B * \log_2(1 + SNR)$$

- **C** = max capacity (bits/s)
- **B** = channel bandwidth (Hz)
- **SNR** = signal power / noise power

Example: classic telephony — $B = 4 \text{ kHz}$, $SNR = 1000 \text{ (30 dB)}$ → $C = 40 \text{ kbit/s}$

1.5 Timeline: Wired Access Technologies

Technology	Downlink	Uplink	Note
ADSL	up to 12 Mbit/s	1.3 Mbit/s	—
ADSL2+	up to 24 Mbit/s	3.5 Mbit/s	—
VDSL2	up to 200 Mbit/s total	—	Belgium: Proximus rollout
G.fast	250-1000 Mbit/s total	—	DSL replacement; <250 m loops
FTTB	G.fast over in-building cabling	—	Fiber to basement
Optical fiber	>1 Gbit/s	—	Only option for Gbit+

Access speeds roughly double every 21 months (**Nielsen's Law**).

2. Optical Fiber Access Technologies

2.1 FTTx: Fiber to the x

Optical fiber gets closer and closer to the user:

```
CO --fiber--> Remote node --copper--> Home    (ADSL/VDSL)
CO --fiber-----> Home    (FTTH)
CO --fiber--> Building basement -----> Home    (FTTB)
```

Acronym	Meaning
FTTx	Fiber to the x (generic)
FTTB	Fiber to the Building/Basement
FTTH	Fiber to the Home
CO	Central Office

Key FTTH components:

Component	Location	Role
OLT (Optical Line Terminal)	Central Office	Sends optical signals; receives upstream from ONUs
Remote Node	Street cabinet	Passive or active switching/splitting
ONU (Optical Network Unit)	Customer premises (CPE)	Terminates optical signal at the home

Optical communication chain:

```
Info source -> Electrical -> [Laser diode] -> Optical fiber -> [Photo diode] -> Electrical -> Info sink
```

FTTH Deployment methods:

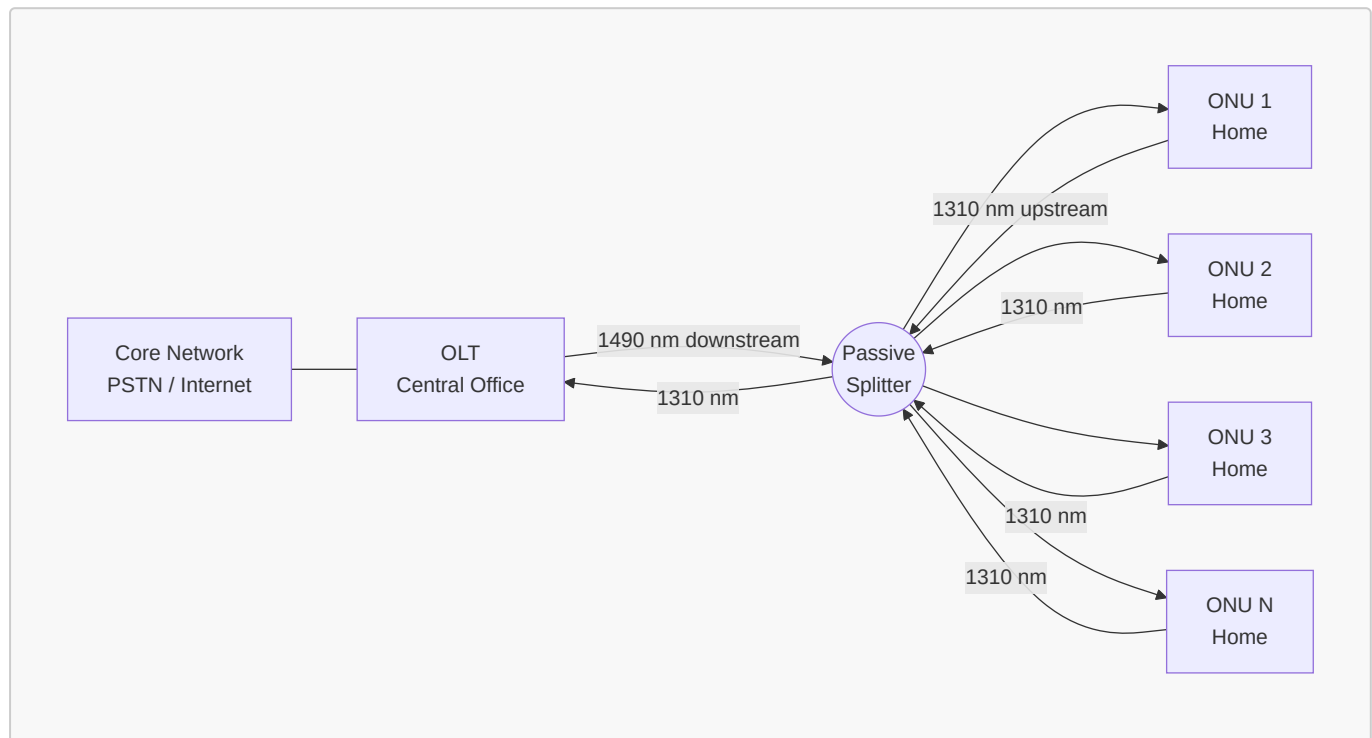
- **Digging:** most common, ~50 EUR/meter; high cost
- **Blowing/pulling:** through pre-installed tubes; cheaper
- **Aboveground:** poles/facades (common in Japan); cheaper but less aesthetic

2.2 FTTH Architectures

Architecture	Remote node	Pros	Cons
Point-to-Point	Patch panel	Dedicated fiber; max privacy	Expensive (many fibers)

Architecture	Remote node	Pros	Cons
AON (Active Optical Network)	Ethernet switch	Flexible	Needs power in street cabinet
PON (Passive Optical Network)	Passive splitter	No power needed; cost-efficient	Shared medium — MAC needed

PON = most widespread FTTH technology



Both wavelengths travel the **same fiber simultaneously** — the WDM diplexer at each end separates them. IPACT is only needed because all ONUs share the **same upstream wavelength**.

3. TDM-based PON

3.1 PON Basics

- Remote node = **passive power splitter** (no active electronics, no power needed)
- Shared medium** between all ONUs
- Uses **2 (or 3) wavelengths**:

Direction	Wavelength	Notes
Upstream	1310 nm	ONU to OLT
Downstream	1490 nm	OLT to ONU
Analog video (optional)	1550 nm	—

- Bit rates: 1 Gbps up to 10 Gbps (shared among all ONUs)

3.2 Downstream Traffic

Downstream = point-to-multipoint:

- OLT broadcasts packets to all ONUs through the splitter
- Each ONU **filters** and keeps only packets addressed to it
- No collision problem (only one sender: the OLT)

3.3 Upstream Traffic**Upstream = multipoint-to-point:**

- Multiple ONUs share a single fiber toward the OLT
- Without coordination there are **collisions** at the splitter
- Solution: **MAC protocol** (TDM) assigning each ONU a time slot

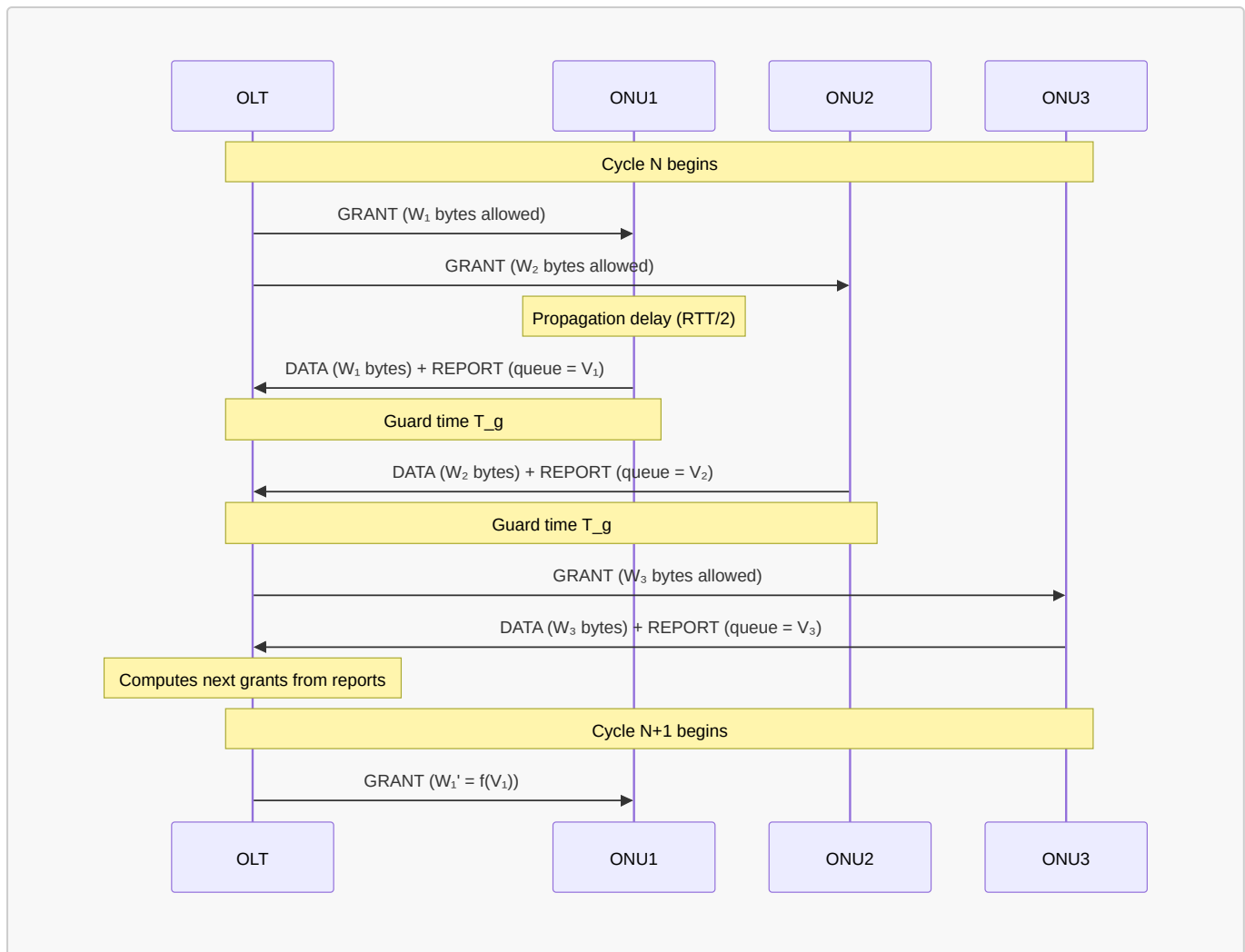
3.4 Bandwidth Allocation

Type	Mechanism	Pros	Cons
Fixed bandwidth allocation	Equal slot size for all ONUs	Simple, fair	Inefficient; wastes bandwidth; higher delay
Dynamic bandwidth allocation (DBA)	Slot size proportional to ONU load	Efficient; lower delay	More complex

4. IPACT — The DBA Protocol for EPON**IPACT = Interleaved Polling with Adaptive Cycle Times****4.1 How IPACT Works**

IPACT operates in a **round-robin** cycle:

1. **OLT sends GRANT** to an ONU — grants permission to send W bytes
2. **ONU sends DATA** (upstream) during its granted window
3. **ONU appends REPORT** at the end of data — tells OLT how many bytes are waiting in its queue
4. **OLT computes next GRANT** based on the REPORT — perfectly times the next slot
5. **Guard band (T_g)** between consecutive ONU transmissions avoids collisions



The OLT **pipelines** grants — it sends GRANT to ONU2 while ONU1's data is still in flight. The grant size $f(V)$ depends on the discipline: Fixed ignores V , Gated sets $W=V$, Limited sets $W=\min(V, W_{max})$.

How does an ONU know when it is allowed to transmit?

The GRANT message contains **two** pieces of information:

1. **Window size** — how many bytes the ONU is allowed to send
2. **Start time (timestamp)** — the exact moment the ONU must begin transmitting

The OLT knows the RTT to every ONU (measured during registration) and uses it to **pre-calculate** exactly when each upstream burst will arrive at the splitter, so slots fit together without colliding:

OLT computes:

"ONU 2's grant travels $RTT/2 = 100 \mu s$ to reach ONU 2.

ONU 1 finishes at time T .

So I tell ONU 2: start transmitting at $T + T_g - RTT/2$,
so its signal arrives at the splitter at exactly $T + T_g$."

On the ONU side:

- Receives the GRANT (on 1490 nm downstream)

- Reads the start timestamp from the GRANT
- Waits until that exact moment
- Fires its upstream burst (on 1310 nm)

Why the guard time $T_g = 1 \mu s$ exists: Timing can never be perfect — slight clock drift, small distance measurement errors, etc. The guard time is a safety margin so no two ONUs ever overlap at the splitter even with minor inaccuracies.

Why ONUs cannot just "listen before transmitting" like Ethernet:

- The splitter is **passive** — it combines signals blindly with no intelligence
- An ONU cannot hear other ONUs' upstream traffic (the upstream signal has already passed the splitter heading toward the OLT before any ONU could detect it)
- Collision avoidance **must be centrally scheduled** by the OLT — not distributed

The OLT is the conductor, the GRANT is the baton cue, and each ONU plays exactly when told.

The cycle length adapts to the total traffic load — hence **adaptive cycle times**.

4.2 Key Parameters (reference scenario)

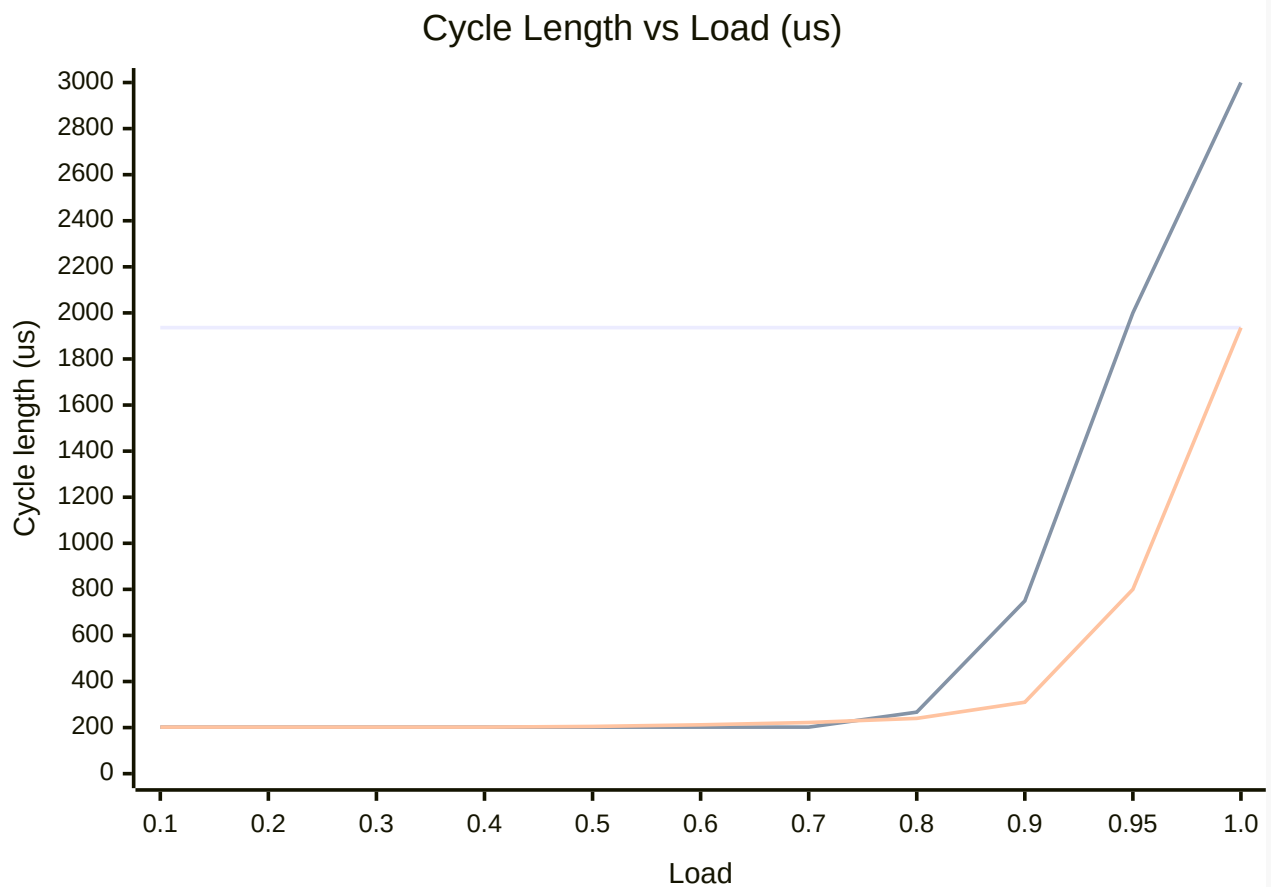
Parameter	Symbol	Value
Number of ONUs	N	16
OLT-ONU distance	d	20 km
Max window size	W_{max}	15,000 bytes
Guard time	T_g	1 μs
Buffer size	—	10 MB
REPORT / GRANT message size	—	64 bytes
Upstream line rate	R	1 Gbps
Refractive index	η	1.5
Speed in fiber	v	200,000 km/s

4.3 IPACT Service Disciplines

The OLT decides how much to grant (W_i) based on the ONU's request (V_i):

Discipline	Grant formula	Behavior
Fixed	$W_i = W_{max}$	Always grants fixed max; ignores REPORT; full fairness
Gated	$W_i = V_i$	Grants exactly what is requested; fully adaptive; can be unfair
Limited	$W_i = \min(V_i, W_{max})$	Compromise: caps grant at W_{max} ; prevents starvation

Cycle length across disciplines and loads:



Charts not fully accurate, for illustration purposes only

Fixed (top, flat 1936 μ s) · **Gated** (middle, RTT-limited until $\sim L=0.8$, then diverges — capped at 3000 μ s, actually $\rightarrow \infty$ at $L=1$) · **Limited** (bottom, grows slowly, converges back to 1936 μ s at $L=1$ since all ONUs always have $W_{\max}$ bytes ready)

4.4 Important Timing Formulas

RTT:

$$RTT = 2 * d / v$$

For $d = 20$ km, $v = 200,000$ km/s: $RTT = \mathbf{200 \mu s}$

Transmission time of $W_{\max}$:

$$T_{W_{\max}} = 15000 * 8 / 10^9 = 120 \mu s$$

Amount of bits / bitrate

Transmission time of REPORT or GRANT (64 bytes):

$$T_{ctrl} = 64 * 8 / 10^9 = 0.512 \text{ us} \sim 0.5 \text{ us}$$

Minimum cycle length (Gated/Limited):

$$T_{cycle_min} = RTT + T_{report} + T_{grant} = 200 + 0.5 + 0.5 = 201 \text{ us}$$

This is the absolute minimum — the OLT must wait for the RTT before it can act on a REPORT.

4.5 IPACT Fixed — Analysis

Cycle length (constant):

$$T_{cycle} = N * (T_{wmax} + T_g) = 16 * (120 + 1) = 1936 \text{ us}$$

Waiting time summary:

Scenario	Waiting time
Minimum	0 (packet arrives exactly when slot begins)
Maximum (low load)	$T_{cycle} - T_{Wmax} = 1936 - 120 = 1816 \text{ us}$
Average (low load)	$(15/16) * 908 = 851 \text{ us}$
Saturation	~1.3 s (buffer fills up)

Throughput:

$$\begin{aligned} \text{Guard overhead} &= N * T_g = 16 * 1 \text{ us} = 16 \text{ us per cycle} \\ \text{Max throughput} &= (1936 - 16) / 1936 * 1 \text{ Gbps} \sim 990 \text{ Mbps} \end{aligned}$$

4.6 IPACT Gated — Analysis

Cycle length (load-dependent) — derivation:

Two equations, one unknown (X = data transmission time per ONU per cycle):

Equation 1 — Cycle structure (N ONUs, each sending data X + overhead per slot):

$$\begin{aligned} T &= N * (X + T_{report} + T_g) \\ T &= 16 * (X + 1.5 \text{ us}) \\ T &= 16X + 24 \text{ us} \end{aligned}$$

Equation 2 — Load definition (L = fraction of upstream capacity used by data):

$$L = (N \times X) / T \quad \rightarrow \quad 16X = L \times T$$

Solve (substitute Eq.2 into Eq.1):

```
T = L×T + 24
T(1 - L) = 24
T = 24 / (1 - L) [us]      <-- correct derivation
(course notes write 24/[L*(1-L)] – close at high load but cannot be derived
from first principles)
```

Apply for $L = 0.1$ (low load):

$$T = 24 / (1 - 0.1) = 24 / 0.9 = 26.7 \text{ us}$$

26.7 us < RTT minimum of 201 us → formula invalid — the cycle can never be shorter than the RTT. Use the physical minimum instead:

$$T = \text{RTT} + T_{\text{report}} + T_{\text{grant}} = 200 + 0.5 + 0.5 = 202 \text{ us}$$

Apply for $L = 0.99$ (high load):

$$T = 24 / (1 - 0.99) = 24 / 0.01 = 2400 \text{ us} \quad (\sim 2424 \text{ us from end results})$$

The 24 us = fixed overhead per cycle (16 ONUs × 1.5 μs each). At high load (1-L is tiny) the denominator shrinks → cycle blows up. At low load the formula gives a cycle shorter than the RTT, which is physically impossible → RTT = 201 μs is the hard floor.

Load	Formula result	Actual cycle
$L = 0.1$	26.7 us (< RTT)	202 us (RTT-limited)
$L = 0.99$	~2400 us	~2424 us

Waiting time:

Load	Average waiting time
Very low ($L \rightarrow 0$)	$1.5 \times 202 = \mathbf{303 \text{ us}}$
High ($L = 0.99$)	~3600 us

Load	Average waiting time
Saturation	≥ 0.96 s

Throughput/Goodput:

- Max throughput: **1 Gbps - 25 kbps**
- Max goodput: **1 Gbps - 37 kbps**

4.7 IPACT Limited — Analysis

Behaves like Fixed at low load (cycle = minimum) and approaches Fixed at high load:

Load	Cycle length	Avg. waiting time
Minimum	201 us	—
L = 0.1	202 us	303 us
L = 0.9	240 us	360 us
Saturation	—	~1.3 s

4.8 IPACT Limited & USRs (Slot Underutilization)

In IPACT Limited, **Ethernet packets cannot be fragmented**. If the granted window is smaller than the head-of-queue packet, that packet **cannot be transmitted** — bandwidth is wasted.

This is called **slot underutilization** or **Unused Slot Remainders (USRs)**.

Maximum slot underutilization:

$$\text{USR}_{\text{max}} = (1518 - 1) / 15000 = 1517 / 15000 \sim 10.1\%$$

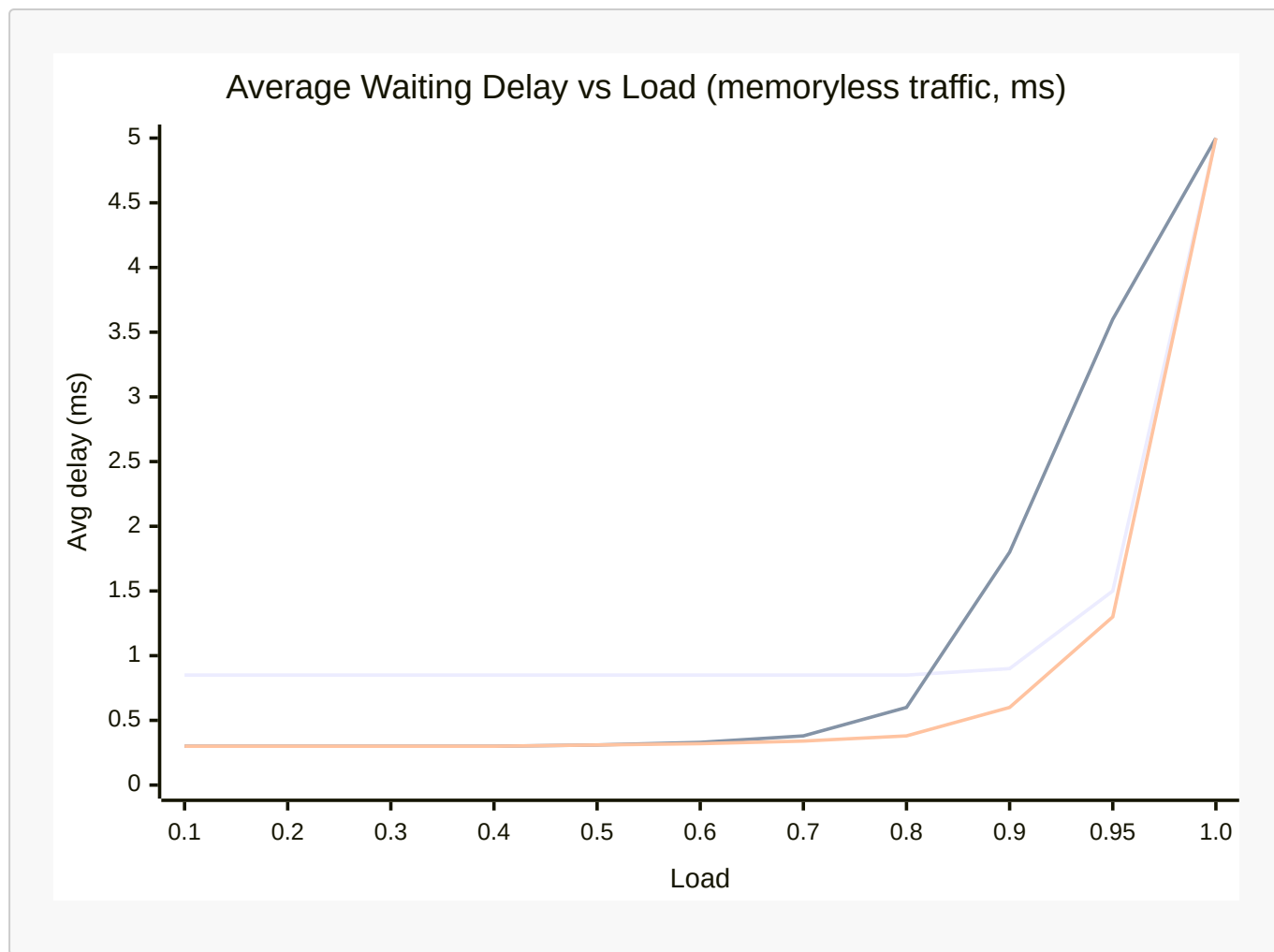
(Worst case: granted slot is 1 byte short of a 1518-byte frame — 1517 bytes wasted)

With 2.5% USR assumed:

- Max throughput: **96.7%**
- Max goodput: **96.3%**

5. IPACT Performance Graphs

5.1 Delay vs. Load Behavior



Fixed (top, flat ~0.85 ms) · **Gated** (middle, low then diverges fastest — already 3.6 ms at L=0.99) · **Limited** (bottom, lowest until near saturation) At L=1.0 all three disciplines diverge to infinity in theory (queues grow without bound). In practice buffers fill → **packets drop** → system stabilises at a lossy operating point. Chart capped at 5 ms; real saturation delays reach seconds.

Fixed:

- Baseline delay ~851 μ s at low load — **flat and high** regardless of load
- Stays flat until load approaches 1, then spikes sharply
- High baseline because cycle length never shrinks even when ONUs have little traffic

Gated:

- Very low delay at low load (~303 μ s) — cycle contracts to minimum
- Delay slowly increases with load as cycle grows
- Near saturation: delay diverges (queues grow unboundedly)

Limited:

- Same low delay as Gated at low load (~303 μ s)
- Stays lower than Gated up to moderate load
- Spikes sharply near saturation (like Fixed) because W_max caps prevent the cycle from shrinking enough

5.2 Key Insights

- **Limited is best at low-to-medium load** (short cycle + fairness)
- **Gated is best at very high load** (no $W_{\max}$ cap — can drain queue faster)
- **Fixed always has the highest baseline delay** but also the most predictable timing
- **Bursty traffic** amplifies the differences between disciplines more than memoryless traffic

6. PON Standards

Standard	DS rate	US rate	Reach	Split ratio
EPON	1.25 Gbps	1.25 Gbps	10-20 km	1:16
GPON	2.5 Gbps	1.25 Gbps	20 km	1:32 to 1:128
10G-EPON	10.3 Gbps	1.25/10.3 Gbps	10-20 km	1:16/32
XG-PON	10 Gbps	2.5/10 Gbps	20 km	1:32 to 1:128

GPON allows packet fragmentation — no USR problem, higher throughput, more flexible cycle lengths.

7. Next-Generation PON

7.1 Motivation

- Higher aggregate bandwidth (≥ 40 Gbit/s per feeder fiber)
- Node consolidation: fewer COs, long-reach PON
- Smooth migration path from TDM-PON

7.2 WDM-PON

Each ONU gets a **dedicated wavelength** (no time sharing, no MAC protocol needed).

Variant	Frequency spacing	Wavelength range	# Wavelengths
CWDM-PON	>200 GHz	1271-1611 nm	18
DWDM-PON	100/50 GHz	C-band (1530-1565 nm)	40-80
UDWDM-PON	12.5 GHz	C-band	320

AWG vs. power splitter at remote node:

- **AWG** (Arrayed Waveguide Grating): fixed 3 dB insertion loss; routes each wavelength to specific ONU
- **Power splitter**: insertion loss = $3 \cdot \log_2(n)$ dB; ONU selects wavelength with tunable filter

7.3 Other NG-PON Technologies

- **TWDM-PON**: Time + Wavelength Division Multiplexing; standard for NGPON-2
- **OFDM-PON**: uses OFDM modulation for flexible spectrum allocation per user

8. Market Evolution

- **Asia** (Japan, South Korea): almost exclusively fiber-based new broadband equipment
- **Europe**: mix of cable, copper (VDSL), and fiber
- **OECD (2021)**: Fibre = 32.1%, Cable modem = 33.8%, DSL = 28.7%
- Large difference between countries in copper vs. fiber penetration (OECD broadband chart)

PART 2: EXERCISES WITH SOLUTIONS

Reference Scenario (all exercises)

Parameter	Value
N (ONUs)	16
d (distance OLT-ONU)	20 km
W_max	15,000 bytes
T_g (guard time)	1 us
Buffer size	10 MB
REPORT / GRANT size	64 bytes
R (upstream line rate)	1 Gbps
Traffic	Uniform load, memoryless

Pre-computed values:

- $v = 300,000 / 1.5 = \mathbf{200,000 \text{ km/s}}$
- $RTT = 2 \times 20 / 200,000 = \mathbf{200 \text{ us}}$
- $T_{Wmax} = 15,000 \times 8 / 10^9 = \mathbf{120 \text{ us}}$
- $T_{report} = T_{grant} = 64 \times 8 / 10^9 = \mathbf{0.5 \text{ us}}$

Exercise 1: IPACT Fixed

a) RTT

$$RTT = 2 * d / v = 2 * 20 / 200,000 = 200 \text{ us}$$

b) Cycle length

Every ONU always sends W_max bytes:

$$T_{cycle} = N * (T_{Wmax} + T_g) = 16 * (120 + 1) = 1936 \text{ us}$$

c) Minimum waiting time

A packet arrives right when the ONU's slot begins — it is transmitted immediately:

$$w_{\min} = 0$$

d) Maximum waiting time at low load ($L \rightarrow 0$)

A packet arrives right after the ONU's slot ends — must wait almost a full cycle:

$$w_{\max} = T_{\text{cycle}} - T_{\text{wmax}} = 1936 - 120 = 1816 \text{ us}$$

e) Average waiting time at low load ($L \rightarrow 0$)

- 1/16 of packets arrive during ONU's own slot: wait = 0
- 15/16 arrive outside: wait uniformly between 0 and 1816 us, avg = 908 us

$$w_{\text{avg}} = (15/16) * 908 = 851 \text{ us}$$

f) Average waiting time at high load ($L \rightarrow \text{infinity}$ / saturation)

Buffer fills completely. Service rate per ONU = $(120/1936) * 1 \text{ Gbps} = 62 \text{ Mbps}$. At saturation, buffer = 10 MB must drain:

$$w_{\text{sat}} \sim \text{buffer} / \text{service_rate} = (10 * 10^6 * 8) / (62 * 10^6) \sim 1.3 \text{ s}$$

g) Maximum burst at 100 Mbps before packet loss

Net fill rate of buffer = $100 - 62 = 38 \text{ Mbps}$:

$$t_{\text{burst}} = (10 * 10^6 * 8) / (38 * 10^6) \sim 2.1 \text{ s}$$

h) Maximum throughput

Guard overhead = $N * T_g = 16 * 1 \text{ us} = 16 \text{ us}$ per cycle (16,000 bits):

$$\text{Throughput}_{\max} = (1936 - 16) / 1936 * 1 \text{ Gbps} \sim 990 \text{ Mbps}$$

i) Maximum goodput

REPORT and GRANT overhead is small (only 16 us total) relative to 1936 us cycle:

Goodput ~ 990 Mbps

j) Understanding the delay vs. load graph for Fixed

- Delay is approximately constant (~851 us) regardless of load until close to saturation
- This is because the cycle length never changes — even idle ONUs still occupy their full slot
- Near load = 1 the delay spikes sharply as queues overflow

Exercise 2: IPACT Gated

a) Minimum cycle length

The OLT must wait for the RTT before acting on a REPORT:

$$T_{\text{cycle_min}} = \text{RTT} + T_{\text{report}} + T_{\text{grant}} = 200 + 0.5 + 0.5 = 201 \text{ us}$$

b) Average cycle length at very low load (L = 0.1)

At very low load each ONU requests almost nothing — cycle is RTT-limited:

$$T_{\text{cycle}} \sim 202 \text{ us}$$

c) Average cycle length at high load (L = 0.99)

Using load formula (valid for higher loads):

$$\begin{aligned} T_{\text{cycle}} &= N * (T_{\text{report}} + T_{\text{g}}) / [L * (1-L)] \\ &= 16 * 1.5 \text{ us} / [0.99 * 0.01] \\ &= 24 / 0.0099 \\ &\sim 2424 \text{ us} \end{aligned}$$

d) Average waiting time at very low load

Waiting time = between 1 and 2 cycle lengths; average = 1.5 * T_cycle:

$$W_{\text{avg}} = 1.5 * 202 = 303 \text{ us}$$

e) Average waiting time at high load (L = 0.99)

$$W_{avg} \sim 3600 \text{ us}$$

(The cycle is ~2424 us; waiting time grows proportionally)

f) Average waiting time at saturation

The derivation links directly to h) (throughput). From h), the guard-time overhead = 25 kbps:

$$\text{overhead_rate} = N * T_g * R / T_{\text{cycle}} = 16 * 1 \text{ us} * 1 \text{ Gbps} / T_{\text{cycle}} = 16,000 \text{ bits} / T_{\text{cycle}}$$

Set equal to 25 kbps to find the saturation cycle:

$$\begin{aligned} 16,000 \text{ bits} / T_{\text{sat}} &= 25,000 \text{ bps} \\ T_{\text{sat}} &= 16,000 / 25,000 \text{ s} = 0.64 \text{ s} = 640 \text{ ms} \end{aligned}$$

Apply the same waiting time formula used in d) (packet waits between 1 and 2 cycles, average 1.5):

$$W_{\text{sat}} \geq 1.5 * T_{\text{sat}} = 1.5 * 640 \text{ ms} = 960 \text{ ms} \geq 0.96 \text{ s}$$

The "≥" is because $T_{\text{sat}} = 640 \text{ ms}$ is a reference cycle — as load pushes further toward 1, T_{cycle} only grows larger so the waiting time can only increase.

h) and f) use the same reference cycle ($T = 640 \text{ ms}$): h) gives the overhead rate at that cycle (25 kbps), and f) gives $1.5 \times$ that cycle as the minimum waiting time (0.96 s). Two sides of the same coin.

g) Average queue filling at 50 Mbps arrival rate

$$\text{Queue fill} = W_{avg} * \text{arrival_rate} = 303 \text{ us} * 50 \text{ Mbps} = 303e-6 * 50e6 = 15.15 \text{ kbit} \sim 16 \text{ kbit}$$

h) Maximum throughput

Overhead = guard times = $16 * 1 \text{ us} = 16 \text{ us}$ per cycle minimum. At max throughput cycle = minimum:

$$\text{Throughput_max} = 1 \text{ Gbps} - 25 \text{ kbps}$$

i) Maximum goodput

Also subtracting REPORT + GRANT:

$$\text{Goodput_max} = 1 \text{ Gbps} - 37 \text{ kbps}$$

j) Understanding the delay vs. load graph for Gated

- At low load: delay is very low (~303 us) because cycle shrinks to minimum
- As load increases, cycle grows and delay grows
- Near saturation: delay diverges (queues grow without bound)
- Much better than Fixed at low-medium load

Exercise 3: IPACT Limited

a) Minimum cycle length

Same as Gated (RTT-limited):

$$T_{\text{cycle_min}} = 201 \text{ us}$$

b) Average cycle length at very low load

$$T_{\text{cycle}} \sim 202 \text{ us}$$

c) Average cycle length at high load ($L = 0.9$)

Limited caps grants at W_{max} but cycle still grows slightly:

$$T_{\text{cycle}} \sim 240 \text{ us}$$

d) Average waiting time at very low load

$$W_{\text{avg}} \sim 303 \text{ us}$$

(Same as Gated at low load — identical short cycle)

e) Average waiting time at high load ($L = 0.9$)

$$W_{\text{avg}} \sim 360 \text{ us}$$

f) Average waiting time at saturation

$$W_{\text{sat}} \sim 1.3 \text{ s}$$

g) Understanding the delay vs. load graph for Limited

- Same low delay as Gated at low load
- Better than Gated at medium load (cycle doesn't grow as fast)
- Worse than Gated near saturation (W_{max} cap prevents efficient draining)
- Summary: Limited is the best choice for low-to-medium load scenarios

Exercise 3bis: IPACT Limited & USRs

a) Maximum % slot underutilization

Worst case: granted slot is 1 byte less than a full 1518-byte Ethernet frame:

$$USR_{\text{max}} = (1518 - 1) / 15000 = 1517 / 15000 \sim 10.1\%$$

b) Maximum throughput assuming 2.5% USR

Guard time overhead: $16 * 1 \text{ us} = 16 \text{ us}$ out of 1936 us cycle:

$$\begin{aligned} \text{Total overhead} &= \text{guard overhead} + \text{USR overhead} \\ (1936 - 16) / 1936 * (1 - 0.025) * 1 \text{ Gbps} &\sim 96.7\% \end{aligned}$$

$$\text{Throughput}_{\text{max}} \sim 96.7\%$$

c) Maximum goodput assuming 2.5% USR

Also subtracting REPORT/GRANT overhead:

$$\text{Goodput}_{\text{max}} \sim 96.3\%$$

Exercise 4: Conceptual Reasoning

4a) Disadvantage of adaptive cycle lengths

No timing guarantees. The OLT cannot predict exactly when a given ONU will get its next slot because the cycle length depends on all ONUs' instantaneous traffic. This makes it hard to guarantee bounded delay or QoS for real-time traffic (voice, video).

4b) If W_{max} increases, does waiting time increase or decrease in Limited?

It **increases** when the load is high enough to fill the larger $W_{\max}$ slots. With larger $W_{\max}$, each ONU transmits more per cycle — the cycle length grows — so the waiting time increases. At very low load the effect is minimal (slots rarely reach $W_{\max}$ anyway).

4c) Why prefer Limited over Gated despite worse performance?

Fairness. In Gated, an ONU with a large backlog receives a proportionally large grant, potentially starving other ONUs for multiple cycles. Limited caps the grant at $W_{\max}$, ensuring all ONUs get service within each cycle — guaranteed max-min fairness and bounded service time.

4d) Credit-based grant sizing (OLT grants more than requested)

Benefits (+):

- Can also send packets that arrived at the ONU **after** the REPORT was sent (during the RTT)
- Reduces wasted cycle time at high constant load
- Better efficiency for continuous/constant traffic

Drawbacks (-):

- Can be pure overhead at low load — unused credit = wasted bandwidth
- Less fair between ONUs

Performance:

- **Better for high, constant load** (credit is always used)
- **Worse for low/bursty load** (credit often goes unused, wasting bandwidth)

4bis-a) Extended reach to 100 km — minimum average waiting time

$$RTT_{100km} = 2 * 100 / 200,000 = 1 \text{ ms}$$

$$T_{\text{cycle_min}} = RTT + T_{\text{report}} + T_{\text{grant}} = 1 \text{ ms} + 0.5 \text{ us} + 0.5 \text{ us} \sim 1.001 \text{ ms}$$

$$W_{\text{avg_min}} = 1.5 * 1.001 \text{ ms} \sim 1.5 \text{ ms}$$

Voice traffic requires max queuing delay < 1 ms. Since RTT alone = 1.001 ms > 1 ms, **IPACT cannot satisfy the 1 ms delay bound at 100 km reach.**

4bis-b) How to improve performance for long reach

- Reduce number of ONUs per OLT — shorter cycle overhead
- Use **credit-based grant sizing** — allows sending packets that arrive during the long RTT, filling otherwise wasted time. Credit-based is **more beneficial** at long reach because the RTT fraction of the cycle is larger.

4bis-c) GPON: effect of packet fragmentation

In GPON packets **can be fragmented** at slot boundaries:

- **No USR problem** — no slot underutilization
- Higher throughput and more flexible cycle lengths
- Throughput and efficiency improve compared to EPON Limited

4bis-d) REPORT transmitted upfront instead of at end

If REPORT is sent at the **beginning** of data:

- OLT receives the REPORT earlier (at start of data slot, not end)
- Next GRANT can be sent sooner — potentially shorter round trip — lower delay
- Disadvantage: the REPORT reflects the queue state at the start of transmission, not the end. It may **undercount** bytes that arrive during transmission — less accurate grant — possible inefficiency
- Also, report is then less up-to-date and small — no real advantage, since other ONUs need their chance to send first

PART 3: SUMMARY TABLES & QUICK REFERENCE

Formula Sheet

Quantity	Formula
Speed in fiber	$v = c / \eta = 300,000 / 1.5 = 200,000 \text{ km/s}$
RTT	$2 * d / v$
T_window	$W_{\text{bytes}} * 8 / R$
Fixed cycle	$N * (T_{W\text{max}} + T_g)$
Gated min cycle	$RTT + T_{\text{report}} + T_{\text{grant}}$
Gated cycle at load L	$N * (T_{\text{report}} + T_g) / [L * (1-L)]$
Avg waiting (low load)	$1.5 * T_{\text{cycle_min}}$
Max throughput	$(T_{\text{cycle}} - \text{overhead}) / T_{\text{cycle}} * R$
USR_max	$(ETH_{\text{max_size}} - 1) / W_{\text{max}}$
Queue fill	$T_{\text{wait}} * \text{arrival_rate}$
Shannon capacity	$B * \log_2(1 + \text{SNR})$

Discipline Comparison

	Fixed	Gated	Limited
Grant formula	$W_i = W_{\text{max}}$	$W_i = V_i$	$W_i = \min(V_i, W_{\text{max}})$
Cycle length	Constant (1936 us)	Variable	Variable (capped)
Low-load delay	High (~851 us)	Low (~303 us)	Low (~303 us)

	Fixed	Gated	Limited
High-load delay	Bounded	Growing	Moderate
Saturation delay	~1.3 s	≥ 0.96 s	~1.3 s
Fairness	Perfect	Poor	Good
Max throughput	~990 Mbps	1 Gbps - 25 kbps	~96.7% (with USR)
USR problem	No	No	Yes
Packet fragmentation	No	No	No

Common Exam Pitfalls

1. **RTT vs. one-way delay:** $RTT = 2 \times \text{propagation delay}$. Always use RTT for min cycle in Gated/Limited.
2. **T_g is one guard band per ONU slot** — not doubled.
3. **Fixed cycle never changes** with load — waiting time is load-independent (only queue depth changes).
4. **Gated cycle CAN be shorter than Fixed:** at low load Gated cycle shrinks to 201 us while Fixed stays at 1936 us.
5. **USR only affects Limited** (not Fixed or Gated) — only Limited can grant a slot smaller than the head-of-queue packet.
6. **Goodput vs Throughput:** Throughput includes REPORT/GRANT overhead; Goodput excludes it.
7. **Waiting time in Fixed at low load is ~851 us, not 0** — even with almost no traffic, a packet must wait for its slot.
8. **Min cycle in Gated can never be less than RTT** — the OLT physically cannot respond to a REPORT before the round-trip time elapses.